



Optimizing emergency services for road safety using a decomposition method: a case study of Delhi

Shayesta Wajid¹ · N. Nezamuddin¹

Accepted: 26 November 2022 / Published online: 12 December 2022

© The Author(s), under exclusive licence to Operational Research Society of India 2022

Abstract

Road traffic crashes are among the top ten leading causes of deaths in India and emergency medical services play a vital role in reducing fatality after a road crash. Being one of the largest metropolitan areas in the world, Delhi has thousands of fatal crashes each year which makes it challenging to achieve a timely response to crashes. In this study, we demonstrated the efficiency of a district-level decomposition approach for solving various types of ambulance optimization problems (for coverage, for survivability, and by incorporating travel time uncertainty) for the city of Delhi, India. Three scenarios were compared to optimize the ambulance locations: the first scenario (*S1*) considered only the existing ambulance sites; the second (*S2*) and third (*S3*) scenarios considered a larger set of potential ambulance sites and relocated ambulances within districts and across districts, respectively. Results showed that the city-level coverage increased by at least 10% with the reallocation of ambulances across districts. For the coverage and survivability maximization models, the decomposition approach was between two to six times faster than the non-decomposition approach. The decomposition approach was critical in solving a robust coverage model accounting for travel time uncertainty and containing nearly half-a-million integer variables. It solved the robust model on a regular PC with 16 GB RAM in half an hour. The non-decomposition approach needed a workstation with 124 GB RAM to solve the same problem in 5 h.

Keywords Emergency medical services · Double standard model · Mixed integer linear programming · Decomposition method · Facility location

✉ N. Nezamuddin
nezam@civil.iitd.ac.in

Shayesta Wajid
shayestawajid@gmail.com

¹ Department of Civil Engineering, Indian Institute of Technology Delhi, New Delhi 110016, India

1 Introduction

The Global Burden of Disease (2019) report ranks road traffic crashes as the world's 7th leading cause of death, holding 1st position for persons aged 10–49 years [1]. In 2016, India faced 2.9% of the deaths due to road traffic injuries, making it the 10th leading cause of death [2]. In Delhi, fatalities due to road traffic crashes have increased from 2.2% in 1990 to 3.1% in 2016. Studies have shown that proper coordination and on-time response of the pre-hospital services can reduce the deaths due to road traffic crashes by approximately 30%. In developing countries like India, 30% of emergency patients die before they reach a hospital. Over 80% of crash victims do not get medical care within 1 h of the incident [1]. Emergency medical service (EMS) systems' performance needs periodic evaluation to monitor their adherence to the standards and reduce the rate of deaths due to unresponsive and delayed emergency services. These prove critical in improving the survival outcome of road traffic crash victims and those in need of pre-hospital care.

The emergency medical services in India are a fragmented system with no set response time standards. Emergency Management and Research Institute (EMRI) runs EMS services in over 15 states in India. States like Maharashtra, Gujarat, Delhi, Bihar, and Haryana have their state government-authorized EMS services. Being run in Public–Private Partnership (PPP) mode poses tremendous difficulty in adhering to particular design and operational standards for such systems' proper functioning. These issues expand the scope of operations research methods for optimizing EMS's strategic, tactical, and operational aspects to save the maximum lives possible [2].

The ambulance location optimization problem for large, congested cities like Delhi, with around 16 million population and thousands of road crash events per year, is computationally burdensome. Hence, this study aims to optimize ambulances' locations across Delhi to maximize the double coverage by decomposing the problem into smaller subproblems. The smaller optimization models are confined to the districts and linked using a single *coupling constraint*. Since our model is an extension of the Double Standard Model (DSM), we call it the *Extended-DSM*.

2 Literature review

The application of facility location problems to the ambulance location domain has been extensive and mostly centred around the coverage problems [3–6]. The first ambulance location model, Location Set Covering Problem (LSCP) minimized the number of ambulances required for coverage, but this number went beyond the budgetary constraints [7]. Addressing this issue, the maximal coverage location problem (MCLP) worked with available ambulances and maximized coverage [8–10]. This model has formed a base to many variants of the

ambulance location models with deterministic [9–11] and probabilistic nature, like the Maximal Expected Covering Location Problem (MEXCLP), Maximal Availability Location Problem and their variants [12–15].

The ambulance unavailability led to incorporation of more ambulances for backup coverage [16, 17]. In this regard, BACOP-1 and BACOP-2 were developed to maximize demand within a single response time standard [18]. Further, the double standard model (DSM) [2] was introduced with a slightly different version of backup coverage considering two response time standards r_1 and r_2 ($r_2 > r_1$). The double standard model ensures coverage of all demand sites within r_2 time units and coverage of a pre-specified proportion (α) of the demand within r_1 time units.

The Double Standard Model has evolved over the years with the addition of multiple parameters [19–31]. One such model restricted the population served by each ambulance [19, 20, 30]. The other served each of the critical and non-critical crash cases by a different vehicle [24, 27]. The multi-period double standard model (m-DSM) [21] and the Robust Double Standard Model (r-DSM) [29, 31] were formulated to incorporate robustness into the model. The m-DSM included complex ambulance relocations across the scenarios, whereas the r-DSM assumed average travel times. Models developed over the last three decades also included uncertainties like travel time, pre-trip delays [32–34], demand uncertainties [35] and multiple time periods [22, 36]. More recent approaches tried to replace the standard 0/1 coverage objective with survival function that redefines the system's nature of pre-hospital care [37–39].

The models were novel in nature but were either applied to small real datasets or test datasets. An increase in the size of the problem leads to a more complex model that becomes computational intractable. Addressing this concern, this study incorporated the *decomposition principle* into the Double Standard Model (DSM) of ambulance location to simultaneously maximize double coverage across all the districts. The district-level sub-problems were linked using a single *coupling constraint* that maintains the total number of ambulances across Delhi.

3 Problem formulation

The traditional DSM [2] works to maximize the demand covered at least twice within a primary coverage standard r_1 using constraints stating that:

- (a) All demand should be covered at least once within secondary coverage standard $r_2 (> r_1)$,
- (b) At least $\alpha \in [0,1]$ fraction of the demand must be covered within the primary coverage standard r_1 ,
- (c) A total of not more than P ambulances should be used,
- (d) Each ambulance site j can have at most u_j ambulances.

According to the emergency services authorities in Delhi, ambulances in each district of Delhi serve the population within that district. To incorporate this operational

feature into our modeling, we developed two new ambulance location models by extending the traditional Double Standard Model. The first model (*Extended-DSM*) decomposes the coverage problem by the districts of Delhi, and the second model (*Robust Extended-DSM*) accounts for travel time variability while maximizing the coverage.

3.1 Extended-DSM

In this model, each district represents an ambulance location problem with demand sites and potential sites specific to that district. Therefore, the ambulance location optimization problem for entire Delhi has been divided into nine subproblems representing one district each. This aspect has been incorporated using the decomposition principle, whereby available ambulances are distributed across nine districts, and multiple instances of DSM representing each district are solved simultaneously. Hence, this model is an extension of DSM and shall be referred to as the *Extended-DSM*.

A schematic diagram of the Extended-DSM is shown in Fig. 1. The variables used in the formulation of the model are given in Table 1.

Formulation of the Extended-DSM is given below.

$$\text{Maximize } f = \sum_{d \in D} \sum_{i=1}^{n_d} w_i^d z_i^d \quad (1)$$

Subject to

$$\sum_{j=1}^{m_d} a_{ij,2}^d x_j^d \geq 1 \quad \forall i \in I, d \in D \quad (2)$$

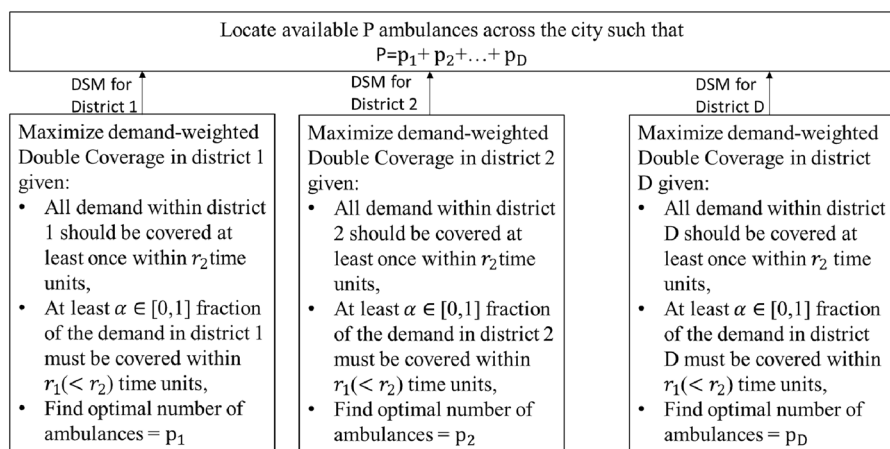


Fig. 1 Example of extended DSM formulation

Table 1 Parameters and variables used in the extended-DSM formulation

Sets	
$i \in I$	Set of call demand locations
$j \in J$	Set of potential ambulance sites
$d \in D$	Set of districts
$s \in S$	Set of demand scenarios
$k \in K$	Set of uncertain travel times in each scenario s
Parameters	
w_i^d	Weight (count of crashes) of each demand node (cluster) i in each district d
t_{ji}^d	Shortest travel time from j to i using historical data in each district d
r_1	Primary coverage standard (minutes)
r_2	Secondary coverage standard (minutes)
α	Level of reliability
P	Total number of ambulances available to be located
p_d	Number of ambulances allocated by CATS system to each district d
u_j	Maximum number of ambulances at potential site j
n_d	Total number of crash sites (demand sites) in each district d
m_d	Total number of potential ambulance locations in each district d
$ D $	Total number of districts
$a_{ij,1}^d$	$= 1$, if $(t_{ji}^s \leq r_1)$: an ambulance at j within district d covers demand at i within r_1 time units and 0 otherwise
$a_{ij,2}^d$	$= 1$, if $(t_{ji}^s \leq r_2)$: an ambulance at j within district d covers demand at i within r_2 time units and 0 otherwise
Decision variables	
z_i^d	$= 1$, if demand node i gets covered twice within r_1 time units in district d , and 0 otherwise
y_i^d	$= 1$, if demand node i gets covered at least once within r_1 time units in district d , and 0 otherwise
x_j^d	Number of vehicles at site j for each district d

$$\sum_{i=1}^{n_d} w_i^d y_i^d \geq \alpha \sum_{i=1}^{n_d} w_i^d \quad \forall d \in D \quad (3)$$

$$\sum_{j=1}^{m_d} a_{ij,1}^d x_j^d \geq z_i^d + y_i^d \quad \forall i \in I, d \in D \quad (4)$$

$$z_i^d \leq y_i^d \quad \forall i \in I, d \in D \quad (5)$$

$$x_j^d \leq u_j \quad \forall j \in J, d \in D \quad (6)$$

$$\sum_{d=1}^D \sum_{j=1}^{m_d} x_j^d \leq P \quad (7)$$

$$y_i^d, z_i^d \in \{0, 1\} \quad \forall i \in I, d \in D \quad (8)$$

$$x_j^d \in \mathbb{Z}^+ \quad \forall j \in J, d \in D \quad (9)$$

The objective function (1) maximizes double coverage of crash locations within the primary coverage standard (r_1) for all the districts combined. Constraint (2) states that all crash locations within each district d must be covered at least once within r_2 time units. According to Constraint (3), at least a portion $\alpha \in [0, 1]$ of the crash sites should be covered within $r_1 (< r_2)$ time units within each district d . Constraint (4) represents the number of ambulances within each district d based on whether crash site i is covered once or twice within r_1 . Constraint (5) states that a demand location i can be double covered only if it is already single covered. Constraint (6) denotes that a maximum of u_j ambulances can be placed at potential site j , and this parameter (u_j) is decided based on ground condition according to CATS authorities.

Constraint (7) states that the total number of ambulances used across all the sites in Delhi cannot exceed the total number of available ambulances. This constraint links all the district-level optimization sub-problems and acts as a *coupling constraint*. Constraint (8) imposes binary constraints on the single and double coverage indicator variables, and constraint (9) restricts the number of ambulances placed at a site to positive integer values.

3.2 Robust extended-DSM

Travel time variability was incorporated into the Extended-DSM (Sect. 3.1) to enable it to provide coverage robustly even during congested traffic conditions. This approach models real-life traffic scenarios faced by the emergency services and builds a model closer to their actual operational setting. The travel time distribution of Delhi was fitted using the Gaussian Mixture Model (GMM) [40].

A set of traffic scenarios (K) was generated by repeatedly drawing travel times from the fitted GMM distribution. Multiple traffic scenarios $k \in K$ were simultaneously incorporated into the Extended-DSM, such that the model provides a feasible solution and maximizes coverage under all traffic conditions. The variables and parameters of the model were indexed using k to denote each traffic scenario. For example, $t_{ji}^{d,k}$ denotes the shortest travel time from j to i in district d in traffic scenario k . The travel time matrices, $a_{ij,1}^d$ and $a_{ij,2}^d$ were further modified to $a_{ij,1}^{d,k}$ and $a_{ij,2}^{d,k}$, respectively for each traffic scenario k . Similarly, the decision variables y_i^d and z_i^d were modified to $y_i^{d,k}$ and $z_i^{d,k}$, respectively, to denote if demand node i gets covered at least once and twice, respectively, within r_1 time units in district d in traffic scenario k . The resulting model, called the *Robust Extended-DSM*, is presented below:

$$\text{Maximize } f = \frac{1}{|K|} \sum_{k \in K} \sum_{d \in D} \sum_{i=1}^{n_d} w_i^d z_i^{d,k} \quad (10)$$

Subject to

$$\sum_{j=1}^{m_d} a_{ij,2}^{d,k} x_j^d \geq 1 \quad \forall i \in I, d \in D, k \in K \quad (11)$$

$$\sum_{i=1}^{n_d} w_i^d y_i^{d,k} \geq \alpha \sum_{i=1}^{n_d} w_i^d \quad \forall d \in D, k \in K \quad (12)$$

$$\sum_{j=1}^{m_d} a_{ij,1}^{d,k} x_j^d \geq z_i^{d,k} + y_i^{d,k} \quad \forall i \in I, d \in D, k \in K \quad (13)$$

$$z_i^{d,k} \leq y_i^{d,k} \quad \forall i \in I, d \in D, k \in K \quad (14)$$

$$x_j^d \leq u_j \quad \forall j \in J, d \in D \quad (15)$$

$$\sum_{d=1}^D \sum_{j=1}^{m_d} x_j^d \leq P \quad (16)$$

$$y_i^{d,k}, z_i^{d,k} \in \{0, 1\} \quad \forall i \in I, d \in D, k \in K \quad (17)$$

$$x_j^d \in \mathbb{Z}^+ \quad \forall j \in J, d \in D \quad (18)$$

The Robust Extended-DSM contains $4|K|$ extra functional constraints. The number of variables and parameters has also increased substantially (due to their indexing with k), which makes the resulting mixed-integer program much more difficult to solve.

4 Case study

The Extended-DSM and Robust Extended-DSM formulated above were applied to a large dataset of Delhi, India. Fatal road traffic crash data were obtained from Delhi Traffic Police for 2014–2016. This data contained fields like date, time and place of crash, offending vehicle, victim and the first information report (FIR) number filed by the police. The traffic crash locations were geocoded and are shown in Fig. 2a.

Delhi is composed of nine districts and the percentage of demand (crashes) in each district is given in Table 2 below. Delhi's city population is over 16 million

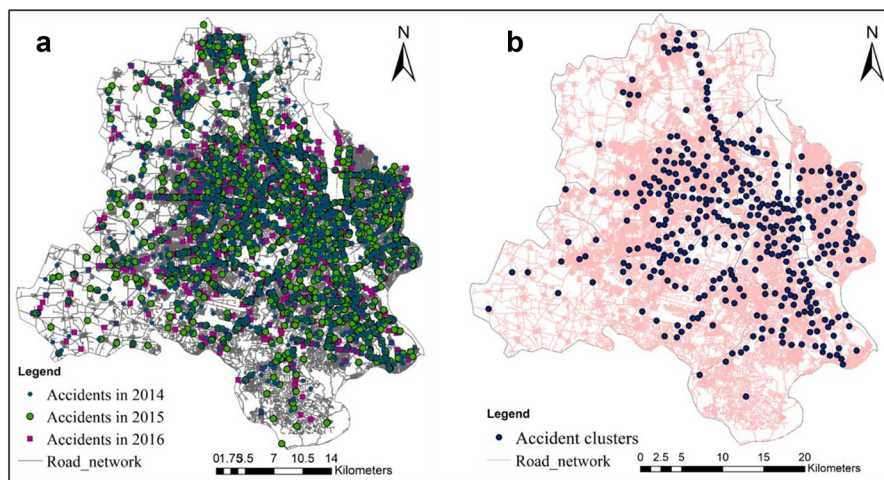


Fig. 2 Distribution of **a** fatal crashes (2014–2016), **b** crash clusters across Delhi

Table 2 Demand per district in Delhi

District	Demand (%)
Central Delhi	4.8
East Delhi	7.5
New Delhi	3.8
North Delhi	10.7
North-East Delhi	7.0
North-West Delhi	24.0
South Delhi	14.1
South-West Delhi	15.8
West Delhi	12.5

and it is one of the largest metropolitan areas in the world. Centralized Accident and Trauma Services (CATS) is a public emergency medical services that has been operational in Delhi since 1991 and works with a fleet of 150 ambulances.

According to an evaluation study of CATS, only 30% of the total calls received by the CATS could be covered in 8 min against the internationally accepted norm of 90% [41]. Besides low coverage values, 28% of the calls go unattended due to ambulance non-availability. Hence, there is a pressing need for redistributing ambulances across the nine districts and optimizing their locations within the districts to enhance the coverage and lower the response time. A detailed description of the inputs required for optimizing Delhi's ambulance location problem is mentioned in the subsequent subsections.

4.1 Demand sites

Road traffic crashes within 500 m were clustered based on the blackspot definition by the Ministry of Road Transport and Highways (MoRTH). A crash blackspot is a 500 m long stretch of road in which either five road crashes (involving fatalities/grievous injuries) or ten fatalities took place during the last three years [42]. Based on this definition, 380 crash clusters were formed as demand points with multiple crash, as shown in Fig. 2b.

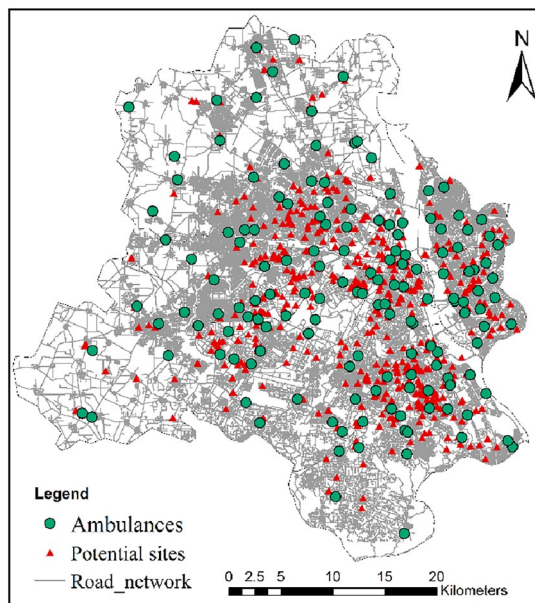
4.2 Potential ambulance locations

To serve 380 demand sites across Delhi, we have identified additional potential ambulance sites and added them to the set of current ambulance locations (Fig. 3). These sites comprise police stations, fire stations and other potential sites that include government buildings and public spaces to station the ambulances. This cumulates to 680 potential sites, including current and potential ambulance locations.

4.3 Travel time computation

For the Extended-DSM, travel times (t_{ji}^d) for the 680×380 Origin (potential ambulance sites)—Destination (crash clusters) matrix were computed using the Distance Matrix API of Google Maps [43, 44]. The API provides travel times for both live

Fig. 3 Distribution of existing and potential ambulance sites across Delhi



traffic conditions and the historical average. Since the model's focus is tactical in nature, historical average travel times were used to optimize ambulance locations.

For the Robust Extended-DSM, to bring robustness into the ambulance location model, another approach incorporating the variability in travel times between the origin–destination pairs was used. Here Maps API was used to capture real-time travel conditions to extract travel time at hourly interval for 24 h for one week [45]. A Gaussian Mixture Model (GMM) with three components was found to be the best fit for Delhi's travel time distribution. For a detailed GMM approach, the readers may refer to [40, 46–48]. Latin Hypercube Sampling approach was used to generate 96 travel time scenarios for each of the 680×380 O-D pairs from the GMM distribution. Now, with the set of demand points (380), potential ambulance sites (680), and two sets of travel time matrices—one based on historical average travel times (680×380) and the other robust ($680 \times 380 \times 96$), we optimized the location of 150 ambulances using the Extended-DSM and Robust Extended-DSM to achieve maximum coverage across all the nine districts.

5 Results and discussion

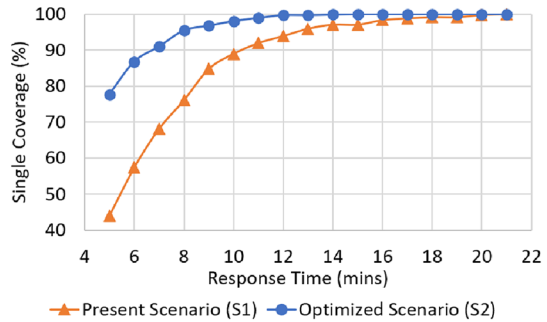
In the absence of response time standard for ambulance operation and to get insights into working of the current system of CATS, this study used the single coverage model [8] to get some idea about the feasible ranges of the parameters of the double coverage model (r_1, r_2, α). The capacity for each ambulance site was set as $u_j = 2$ and all computations were done using Gurobi Optimizer v.8.1.1. Three scenarios were considered to evaluate the differences attained with optimization.

- (1) *S1* (current condition): considered the existing 150 CATS ambulances operating from their current locations (near accident hotspots) in Delhi [49].
- (2) *S2* (district-level optimization): considered all the 680 potential sites to optimize the emergency services. The number of ambulances per district was the same as the CATS authorities provided.
- (3) *S3* (city-level optimization): the model recalculated the number of ambulances for each district, subject to the total number of ambulances being the same for the entire city. All nine subproblems, one for each district, were linked using a *coupling constraint* (7).

5.1 Response time

Figure 4 compares the existing (*S1*) and optimized (*S2*) scenarios for different response times. It took 21 min to cover all crash sites once with the current location of ambulances. But, when the model relocates them considering the potential sites within the district (*S2*), the response time reduces significantly to 15 min. The difference in coverage observed between the two scenarios at 5 min standard is around 30%.

Fig. 4 Comparison of single coverage in current (*S1*) and optimized (*S2*) scenarios for multiple response time standards



A significant difference in the response times for 100% single coverage can also be observed at the district level and for the whole city of Delhi in Fig. 5. The Delhi South, South West, West and North districts reduced response time by more than 5 min with the optimized system. Overall, Delhi showed a 6 min decline in response time with the optimized system.

5.2 Double coverage

The extended double standard model maximizes the double coverage within the primary coverage standard (r_1) for each of the nine districts given the constraints. A difference in double coverage values obtained by varying primary coverage standard (r_1) and reliability value (α) for the existing system (*S1*) and optimized systems (*S2* and *S3*) is shown in Table 3. At $\alpha=0.95$, the existing system offers infeasibility for response times under 13 min. Whereas the optimized systems (*S2* and *S3*) provide feasible double coverage values from 11 min onwards. When α reliability reduces to 0.80, the existing system turns feasible at 11 min and gives 92% double coverage. The optimized systems provide a feasible 97% double coverage at 10 min but show a marginal improvement in coverage values after 10 min.

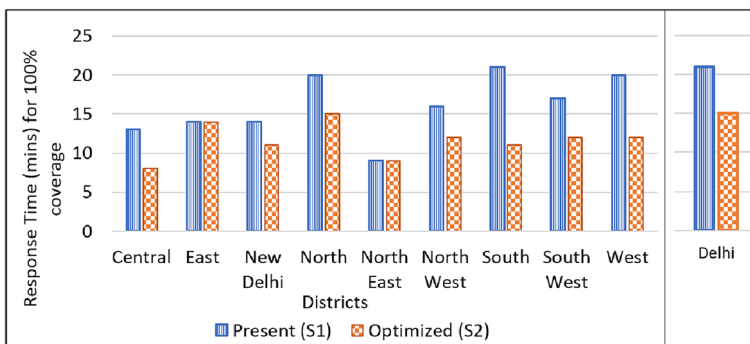


Fig. 5 Response time for each district to achieve 100% single coverage in existing (*S1*) and optimized (*S2*) conditions

Table 3 Double coverage (%) for different r_1 and α values for existing ($S1$) and optimized systems ($S2$ and $S3$) at $r_2=21$ min

r_1	$\alpha=0.8$			$\alpha=0.95$		
	$S1$	$S2$	$S3$	$S1$	$S2$	$S3$
10	Infeasible	97.5	98.1	Infeasible	Infeasible	Infeasible
11	91.9	98.7	98.9	Infeasible	98.7	98.9
12	93.8	99.6	99.7	Infeasible	99.6	99.7
13	95.9	99.8	99.8	Infeasible	99.8	99.8
14	97	99.8	99.8	97	99.8	99.8
15	97	99.8	99.8	97	99.8	99.8

Figure 6 shows the variation in double coverage with an increase in the number of ambulances for the optimized condition at 14 min primary coverage standard. The existing condition ($S1$) represents the maximum coverage (97%) obtained with 150 ambulances at the same response time standard and forms a base case for comparison. From the figure, the optimized system ($S3$) can doubly cover 98% of the demand sites within 14 min with just 70 ambulances. Following this, there is a marginal increase in double coverage with the number of ambulances.

A comparison in double coverage of all three scenarios ($S1$, $S2$, and $S3$) for each district is made for $r_1=13$ min, $r_2=21$ min and $\alpha=0.90$ (Fig. 7). Districts like New Delhi, North, South and Southwest Delhi show 6%, 9%, 2% and 9% differences in double coverage values between existing ($S1$) and optimized scenarios ($S2$ and $S3$).

Fig. 6 Variation in coverage with varying ambulances
 $r_1=14$ min, $r_2=21$ min,
 $\alpha=0.90$

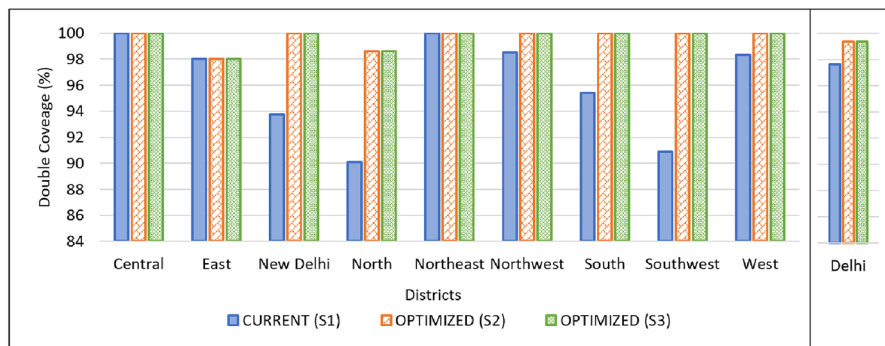
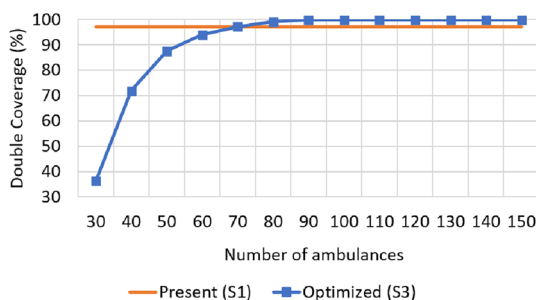


Fig. 7 Comparison of double coverage in each district for $r_1=13$ min, $r_2=21$ min and $\alpha=0.90$

There is no significant difference observed between $S2$ and $S3$ scenarios due to the higher response time standard. The overall model for Delhi shows a 2% rise in coverage from existing to optimized scenarios.

Figure 8 shows the difference between the two optimized scenarios, $S2$ and $S3$, at a primary coverage standard of $r_1=8$ min. Scenario $S3$ computes the number of ambulances in each district for maximizing the coverage subject to the coupling constraint. As a result, ambulances are relocated to districts with higher demands. District-level coverages improved between 0% (East district) to 30% (Southwest district) in $S3$. Overall, Delhi showed a significant 10% increase in coverage after the city-level optimization ($S3$).

5.3 Comparison of decomposition versus non-decomposition approaches

The decomposition approach divides the entire problem for Delhi into multiple sub-problems, each associated with one district of Delhi. The non-decomposition approach does not decompose the problem geographically and solves the entire problem for Delhi at once. The decomposition approach is a clever divide-and-rule approach of solving the problem, whose advantage over the non-decomposition approach depends upon the *size* and *complexity* of the problem being solved. Next, we demonstrate results for one case each of increased problem size and problem complexity for illustration.

5.3.1 Increasing problem size

The first case involved solving the Extended-DSM coverage problem for a larger 1120×1777 O-D matrix taken from another EMS study of Delhi [48]. Comparison of run times and objective function values for the two approaches are shown in Table 4. The problems were solved for four scenarios by varying the number of available ambulances. The objective function values obtained from the two approaches are nearly equal (difference of 0.24%). The decomposition approach

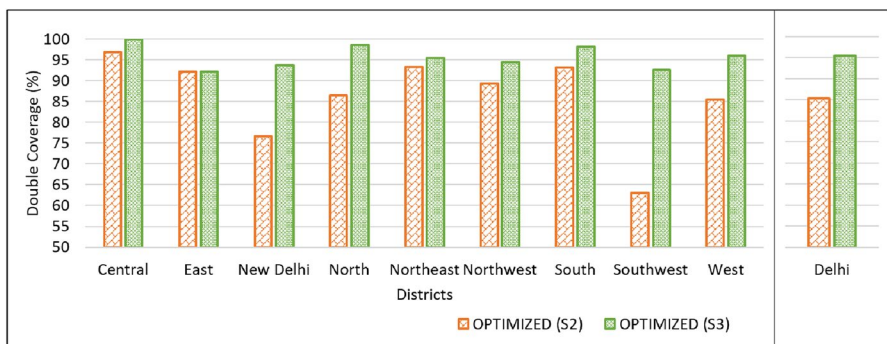


Fig. 8 Comparison of coverage between $S2$ and $S3$ scenarios in each district for $r_1=8$ min, $r_2=15$ min, and $\alpha=0.9$

Table 4 Comparison of runtimes for a larger dataset of O-D pairs

Ambulances	Decomposition		Non-decomposition	
	Runtime (s)	Objective Function value	Runtime (s)	Objective function value
150	40	91,860	287	92,077
259	49	91,860	288	92,077
500	46	91,860	280	92,077
1000	46	91,860	265	92,077

solved nine district-level problems concurrently and was about six times faster than the non-decomposition approach which solved a single problem for entire Delhi.

While absolute run times are of clear interest while solving specific instances of a problem, in algorithmic runtime complexity analysis the rate at which the solution time grows with respect to the problem size is critically important. Therefore, any relative gain in solution time is important as well. The non-decomposition runtime will further increase with respect to the decomposition approach if the problem size increases by increasing the number of available ambulances (by including private EMS operators), increasing the size of the O-D matrix (by clustering the demand sites at a more disaggregate level and/or including more potential ambulance sites), and by including other categories (emergency, high, medium, and low) or types (head injury, chest pain, cardiac arrest, pregnancy, etc.) of calls received by CATS.

5.3.2 Increasing problem complexity

The second case involved solving a Double Standard Robust Survival Model (DSRSM) for efficient utilization of ambulances in Delhi to maximize the survivability of high priority calls received by CATS under uncertain travel times [40]. This model is a mixed integer non-linear problem with higher complexity. The DSRSM model was solved for 680×380 O-D pairs with 150 ambulances, and it was given a threshold runtime of 8 h. In 8 h, the non-decomposition approach obtained an objective function value of 3393.37. The same objective function value was obtained by the decomposition approach in about 4 h, making it twice as fast compared to the non-decomposition approach.

5.4 Difference in performance with robust travel time modeling

This section shows the performance of the Decomposition and Non-decomposition approaches with robust travel time modelling using the Robust Extended-DSM for a larger problem size of 1120×1777 O-D pairs [48]. The robust model was solved using Gurobi v9.1 in x64-based PC with Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz, 6 Core(s) and 12 Logical Processors with 16 GB RAM.

The Robust Extended-DSM for 1120×1777 O-D pairs has 0.43 million integer variables. With 16 GB RAM, which is found in high-end consumer grade laptops/

PCs, the Non-decomposition approach led to memory shortage and was unable to solve the problem. It was eventually solved on a high-performance workstation with 128 GB RAM in approximately 5 h.

The Decomposition approach for the Robust Extended-DSM, on the other hand, solved the entire problem for Delhi under 1800 s (0.5 h) on a regular PC with 16 GB RAM. The results of the two approaches are summarized in Table 5. Not only was the Decomposition approach able to solve the robust model ten times faster, but it also used eight times less memory and can be run on regular laptops/PCs available to analysts and researchers.

6 Conclusion

This study demonstrated the efficiency of a district-level decomposition approach for solving various types of ambulance location optimization models (for coverage, for survivability, and by incorporating travel time uncertainty) for the city of Delhi, India.

Fatal traffic crash data for 2014–2016 were collected from Delhi Traffic Police and clustered to form 380 demand sites. A total of 680 potential ambulance sites were identified to optimally place 150 ambulances of CATS, the public emergency service in Delhi. Maps services (Google Maps and Here Maps APIs) were used to obtain both historical average travel times and live travel times (to model travel time distribution) between ambulance locations and crash clusters O-D pairs.

The single coverage model results indicated that the optimized system takes 6 min less than the current system to cover all crash sites. The double coverage model showed a significant rise in coverage when the ambulances were placed optimally across the city, providing 98% coverage by using only 130 out of the 150 available ambulances. The results also showed that the city-level coverage could be better by at least 10% with the reallocation of ambulances across the nine districts.

Next, we compared the performances of the Decomposition and Non-decomposition approaches to solving various types of ambulance location optimization

Table 5 Non-decomposition versus decomposition for a 1120×1777 dataset

Response time (min)	Non-decomposition (Workstation with 128 GB RAM)		Decomposition (PC with 16 GB RAM)		Gap (%)
	Runtime (s)	Objective function value	Runtime (s)	Objective function value	
9	17,532	93,764.03	1782	93,636.7	0.13
10	17,137	93,897	1605	93,786	0.12
11	17,313	93,931.6	1590	93,835	0.10
12	17,145	93,939.6	1564	93,864	0.09
13	17,086	93,948	1570	93,898.4	0.09
14	16,897	93,963.4	1644	93,938	0.06
15	18,253	93,997	1627	93,972	0.03

models. For the coverage maximization model, the Decomposition approach solved the problem about six times faster than the Non-decomposition approach (40–49 s vs. 265–280 s). For the non-linear survivability maximization model, the Decomposition approach solved the problem about twice as fast (4 vs. 8 h).

The biggest advantage of the Decomposition approach was in solving for a robust coverage model which ensured double coverage to crash sites under travel time uncertainty representing multiple traffic scenarios. The Decomposition approach solved the problem under 1800 s (0.5 h) on a computer with 16 GB RAM. The Non-decomposition approach failed to solve the robust coverage model using 16 GB RAM and ran into memory shortage. Eventually, the Non-decomposition approach was implemented on a workstation with 124 GB RAM, and it took about ten times more to solve the model (5 h). The Decomposition approach made it possible to solve the hard mixed-integer linear robust coverage model with half-a-million integer variables on regular laptops/PCs available to researchers and analysts.

A constraint on the current operation of the emergency services in Delhi is that the ambulances are stationed near accident blackspots for most parts of the year, as per CATS authorities. This reduces the response time of the ambulances for city-wide coverage of road traffic crashes and other emergencies. The problem arises when these locations are kept constant for a long period, as done currently. This problem can be addressed in a future extension of this work that relocates the ambulances according to varying demand pattern. Demand clusters can be updated at some regular interval based on the recent data, and the ambulances can be optimally relocated after running the ambulance optimization models.

Author contributions Data collection, preparation and analysis were performed by SW. The first draft of the manuscript was written by SW. NN commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding No specific funding was received for conducting this study.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability The code used during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflict of interests to declare that are relevant to the content of this article.

Ethics approval Ethics approval was not needed for this study.

References

1. Fitzgerald, M., Dewan, Y., Reilly, G.O., Mathew, J., McKenna, C.: India and the management of road crashes: towards a national trauma system. *Indian J. Surg.* **68**, 226–232 (2006)

2. Gendreau, M., Laporte, G., Semet, F.: Solving an ambulance location model by tabu search. *Locat. Sci.* **5**, 75–88 (1997). [https://doi.org/10.1016/S0966-8349\(97\)00015-6](https://doi.org/10.1016/S0966-8349(97)00015-6)
3. Aringhieri, R., Bruni, M.E., Khodaparasti, S., van Essen, J.T.: Emergency medical services and beyond: addressing new challenges through a wide literature review. *Comput. Oper. Res.* **78**, 349–368 (2017). <https://doi.org/10.1016/j.cor.2016.09.016>
4. Bélanger, V., Ruiz, A., Soriano, P.: Recent optimization models and trends in location, relocation, and dispatching of emergency medical vehicles. *Eur. J. Oper. Res.* **272**, 1–23 (2018). <https://doi.org/10.1016/j.ejor.2018.02.055>
5. Brotcorne, L., Laporte, G., Semet, F.: Ambulance location and relocation models. *Eur. J. Oper. Res.* **147**, 451–463 (2003). [https://doi.org/10.1016/S0377-2217\(02\)00364-8](https://doi.org/10.1016/S0377-2217(02)00364-8)
6. Marianov, V., ReVelle, C.: Siting emergency services. In: Drezner, Z. (ed.) *Facility Location: A Survey of Applications and Methods*, pp. 199–223. Springer, New York (1995)
7. Toregas, C., Swain, R., ReVelle, C., Bergman, L.: The location of emergency service facilities. *Oper. Res.* **19**, 1363–1373 (1971)
8. Church, R., ReVelle, C.: The maximal covering location problem. *Pap. Reg. Sci. Assoc.* **32**, 101–118 (1974). <https://doi.org/10.1007/BF01942293>
9. Eaton, D.J., Daskin, M.S., Simmons, D., Bulloch, B., Jansma, G.: Determining emergency medical service vehicle deployment in Austin, Texas. *Interfaces (Providence)* **15**, 96–108 (1985)
10. Galvão, R.D., ReVelle, C.: A Lagrangean heuristic for the maximal covering location problem. *Eur. J. Oper. Res.* **88**, 114–123 (1996). [https://doi.org/10.1016/0377-2217\(94\)00159-6](https://doi.org/10.1016/0377-2217(94)00159-6)
11. Fitzsimmons, J.A.: A methodology for emergency ambulance deployment. *Manage. Sci.* **19**, 627–636 (1973). <https://doi.org/10.1287/mnsc.19.6.627>
12. Daskin, M.S.: A maximum expected covering location model: formulation, properties and heuristic solution. *Transp. Sci.* **17**, 48–70 (1983)
13. Fujiwara, O., Makjarnroen, T., Gupta, K.K.: Ambulance deployment analysis: a case study of Bangkok. *Eur. J. Oper. Res.* **31**, 9–18 (1987)
14. ReVelle, C., Hogan, K.: The maximum availability location problem. *Transp. Sci.* **23**, 192–200 (1989)
15. McLay, L.A.: A maximum expected covering location model with two types of servers. *IIE Trans. (Inst. Ind. Eng.)* **41**, 730–741 (2009). <https://doi.org/10.1080/07408170802702138>
16. Daskin, M.S., Stern, E.H.: A hierarchical objective set covering model for emergency medical service vehicle deployment. *Transp. Sci.* **15**, 137–152 (1981). <https://doi.org/10.1287/trsc.15.2.137>
17. Eaton, D.J., Sanchez, H.M., Lantigua, R.R., Morgan, J.: Determining ambulance deployment in Santo Domingo, Dominican Republic. *J. Oper. Res. Soc.* **37**, 113–126 (1986)
18. Hogan, K., ReVelle, C.: Concepts and applications of backup coverage. *Manag. Sci.* **32**, 1434–1444 (1986)
19. Doerner, K.F., Gutjahr, W.J., Hartl, R.F., Karall, M., Reimann, M.: Heuristic solution of an extended double-coverage ambulance location problem for Austria. *Cent. Eur. J. Oper. Res.* **13**, 325–340 (2005)
20. Laporte, G., Louveaux, F.V., Semet, F., Thirion, A.: Applications of the double standard model for ambulance location. *Innov. Distrib. Logist.* **619**, 235–249 (2009). <https://doi.org/10.1007/978-3-540-92944-4>
21. Schmid, V., Doerner, K.F.: Ambulance location and relocation problems with time-dependent travel times. *Eur. J. Oper. Res.* **207**, 1293–1303 (2010). <https://doi.org/10.1016/j.ejor.2010.06.033>
22. Başar, A., Çatay, B., Ünlüyurt, T.: A multi-period double coverage approach for locating the emergency medical service stations in Istanbul. *J. Oper. Res. Soc.* **62**, 627–637 (2011). <https://doi.org/10.1057/jors.2010.5>
23. Kepaptsoglou, K., Karlaftis, M.G., Mintsis, G.: Model for planning emergency response services in road safety. *J. Urban Plan. Dev.* **138**, 18–25 (2012). [https://doi.org/10.1061/\(ASCE\)UP.1943-5444.0000097](https://doi.org/10.1061/(ASCE)UP.1943-5444.0000097)
24. Liu, Y., Roshandeh, A.M., Li, Z., Kepaptsoglou, K., Patel, H., Lu, X.: Heuristic approach for optimizing emergency medical services in road safety within large urban networks. *J. Transp. Eng.* **140**, 1–9 (2014). [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000684](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000684)
25. Su, Q., Luo, Q., Huang, S.H.: Cost-effective analyses for emergency medical services deployment: a case study in Shanghai. *Int. J. Prod. Econ.* **163**, 112–123 (2015). <https://doi.org/10.1016/j.ijpe.2015.02.015>

26. Degel, D., Wiesche, L., Rachuba, S., Werners, B.: Time-dependent ambulance allocation considering data-driven empirically required coverage. *Health Care Manag. Sci.* **18**, 444–458 (2015). <https://doi.org/10.1007/s10729-014-9271-5>
27. Liu, Y., Li, Z., Liu, J., Patel, H.: A double standard model for allocating limited emergency medical service vehicle resources ensuring service reliability. *Transp. Res. Part C Emerg. Technol.* **69**, 120–133 (2016). <https://doi.org/10.1016/j.trc.2016.05.023>
28. Liu, M., Yang, D., Hao, F.: Optimization for the locations of ambulances under two-stage life rescue in the emergency medical service: a case study in Shanghai. *China. Math. Probl. Eng.* (2017). <https://doi.org/10.1155/2017/1830480>
29. Dibene, J.C., Maldonado, Y., Vera, C., de Oliveira, M., Trujillo, L., Schütze, O.: Optimizing the location of ambulances in Tijuana, Mexico. *Comput. Biol. Med.* **80**, 107–113 (2017). <https://doi.org/10.1016/j.combiomed.2016.11.016>
30. Mullaoglu, G., Sariyer, G.: Variations of double coverage ambulance location model using call volume and population data. *J. Ind. Prod. Eng.* **36**, 533–545 (2019). <https://doi.org/10.1080/21681015.2019.1701107>
31. Trujillo, L., Álvarez-Hernández, G., Maldonado, Y., Vera, C.: Comparative analysis of relocation strategies for ambulances in the city of Tijuana, Mexico. *Comput. Biol. Med.* **116**, 103567 (2020). <https://doi.org/10.1016/j.combiomed.2019.103567>
32. Ingolfsson, A., Budge, S., Erkut, E.: Optimal ambulance location with random delays and travel times. *Health Care Manag. Sci.* **11**, 262–274 (2008). <https://doi.org/10.1007/s10729-007-9048-1>
33. Goldberg, J., Dietrich, R., Chen, J.M., Mitwasi, M.G., Valenzuela, T., Criss, E.: Validating and applying a model for locating emergency medical vehicles in Tucson, AZ. *Eur. J. Oper. Res.* **49**, 308–324 (1990)
34. van den Berg, P.L., Kommer, G.J., Zuzáková, B.: Linear formulation for the maximum expected coverage location model with fractional coverage. *Oper. Res. Health Care* **8**, 33–41 (2016). <https://doi.org/10.1016/j.orhc.2015.08.001>
35. Ball, M., Lin, F.L.: A reliability model applied to emergency service vehicle location. *Oper. Res.* **41**, 18–36 (1993)
36. Repede, J.F., Bernardo, J.J.: Developing and validating a decision support system for locating emergency medical vehicles in Louisville, Kentucky. *Eur. J. Oper. Res.* **75**, 567–581 (1994)
37. Erkut, E., Armann, I., Erdogan, G.: Ambulance location for maximum survival. *Nav. Res. Logist.* **55**, 541–550 (2007). <https://doi.org/10.1002/nav>
38. Knight, V.A., Harper, P.R., Smith, L.: Ambulance allocation for maximal survival with heterogeneous outcome measures. *Omega* **40**, 918–926 (2012). <https://doi.org/10.1016/j.omega.2012.02.003>
39. Leknes, H., Aartun, E.S., Andersson, H., Christiansen, M., Granberg, T.A.: Strategic ambulance location for heterogeneous regions. *Eur. J. Oper. Res.* **260**, 122–133 (2017). <https://doi.org/10.1016/j.ejor.2016.12.020>
40. Wajid, S., Nezamuddin, N.: A robust survival model for emergency medical services in Delhi, India. *Socio-Econ. Plann. Sci.* (2022). <https://doi.org/10.1016/j.seps.2022.101342>
41. Singh, A.R., Gupta, S.K.: Review of the centralized accident and trauma services (CATS) in New Delhi, India. *Cibtech J. Bio-Protocols* **3**, 45–49 (2014)
42. MoRTH: Protocols for identification and rectification of road accident black spots on national highways (2015)
43. Wang, F., Xu, Y.: Estimating O-D travel time matrix by Google Maps API: implementation, advantages, and implications. *Ann. GIS.* **17**, 199–209 (2011). <https://doi.org/10.1080/19475683.2011.625977>
44. Google: Google maps distance matrix API. <https://developers.google.com/maps/documentation/javascript/distancematrix>
45. HereTechnologies: HERE maps routing API. <https://route.api.here.com/routing/7.2/calculateroute.json>
46. Guo, F., Li, Q., Rakha, H.: Multistate travel time reliability models with skewed component distributions. *Transp. Res. Rec. J. Transp. Res. Board.* **2315**, 47–53 (2012). <https://doi.org/10.3141/2315-05>
47. Kim, S.H., Chung, J.: Exploration on origin–destination-based travel time variability: insights from Seoul metropolitan area. *J. Transp. Geogr.* **70**, 104–113 (2018). <https://doi.org/10.1016/j.jtrangeo.2018.05.021>
48. Wajid, S., Nezamuddin, N.: Capturing delays in response of emergency services in Delhi. Submitted to SEPS, second revision (2021)

49. Govt. of NCT of Delhi.: Centralized Accident and Trauma Services: Ambulance Location Points. http://web.delhi.gov.in/wps/wcm/connect/doiit_cats/CATS/Home/Ambulance+Location+Points

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.